📐 Geometry & Mensuration — Dimension Change Relations

***(Height, Radius, Length, Breadth, Side Changes)* — SSC CGL · CHSL · CPO · Banking · Railways · CAT**

> **The single most predictable scoring topic in the paper.** Once you know *which power* a dimension carries, you never re-derive a formula — you just scale it. This chapter makes every "if side doubles / radius halves / height tripled" question a **5-second mental sum**.

> **Colour legend** (used throughout): 🔵 **Original formula** · 🟢 **New formula** · 🔴 **Shortcut** · 🟠 **Common mistake**

> Companion to Geometry-Handbook-Formulas-and-Figures.md. View the **PDF** so diagrams stay aligned.

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0. The One Idea That Runs the Whole Chapter

Every measurement in geometry is built from powers of length. **Count the powers, and you know the scaling.**

```
┌──────────────────────────────────────────────────────┐
│  QUANTITY          POWER of length      Scale by k means  │
│  ──────────────    ──────────────────   ──────────────── │
│  Length / Perimeter / Circumference /                │
│  Diagonal / Radius / Arc / Slant     →   ¹    →   × k      │
│                                                      │
│  AREA / CSA / LSA / TSA / Surface    →   ²    →   × k²     │
│                                                      │
│  VOLUME                              →   ³    →   × k³     │
└──────────────────────────────────────────────────────┘
```

🔴 **Master shortcut:** If **every** linear dimension is multiplied by *k*:

Quantity	Multiplies by
Length-type (perimeter, diagonal, circumference, arc, slant)	**k**
Area-type (any area, CSA, LSA, TSA)	**k^2**
Volume	**k^3**

Percentage form: if a dimension changes by *x*%, put `k = (1 ± x/100)`, then apply the power.

- Increase 20% → k = 1.2 · Decrease 25% → k = 0.75 · Increase 50% → k = 1.5 · Decrease 20% → k = 0.8

🟠 **The #1 mistake in the whole topic:** applying the linear percentage to area/volume directly. "Side up 20% → area up 20%" is **WRONG**. Area up = $(1.2)^2 - 1$ = **44%**. Always raise to the power *first*, subtract 1 *after*.

0.1 Percentage-Change Ready Reckoner

Keep these `k` and `k²`, `k³` values on your fingertips — most SSC questions use them.

Change	k	k^2 (area)	k^3 (volume)
×2 (double, +100%)	2	4	8
×3 (triple)	3	9	27
÷2 (half, −50%)	0.5	0.25	0.125
+10%	1.1	1.21 (+21%)	1.331 (+33.1%)
+20%	1.2	1.44 (+44%)	1.728 (+72.8%)
+25%	1.25	1.5625 (+56.25%)	1.953 (+95.3%)
+50%	1.5	2.25 (+125%)	3.375 (+237.5%)
−10%	0.9	0.81 (−19%)	0.729 (−27.1%)
−20%	0.8	0.64 (−36%)	0.512 (−48.8%)
−25%	0.75	0.5625 (−43.75%)	0.4219 (−57.8%)

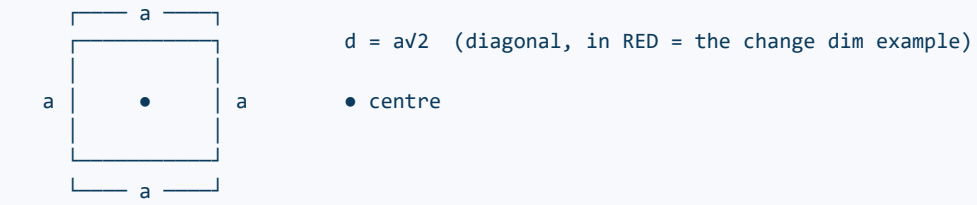
🔴 **Two-factor percentage shortcut (for area / two dimensions):** if one dimension changes by *a*% and another by *b*%, net change = `a + b + (a·b)/100` percent.

Example: length +20%, breadth −20% → `20 − 20 + (20·−20)/100 = −4%` (area **falls 4%**, never 0%).

PART A — 2D FIGURES

*In 2D: perimeter/diagonal/circumference/arc scale as **k**; area scales as **k^2**.*

A1. Square



1. **Original formulas** — 🔵 Area `A = a²` · Perimeter `P = 4a` · Diagonal `d = a√2`
2. **One dimension changes** (a square has only one dimension, the side *a*):

- Side ×2 → 🟢 `A = (2a)² = 4a²` → **Area ×4**; `P = 4(2a) = 8a` → **Perimeter ×2**; diagonal ×2.
- Side ÷2 → Area ×¼, Perimeter ×½, Diagonal ×½.

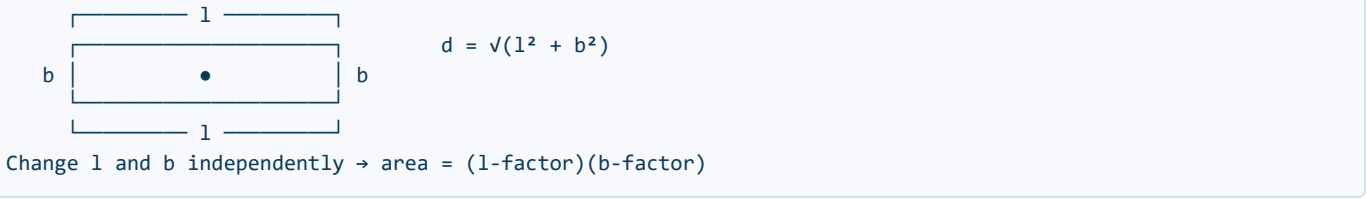
1. **Multiple dimensions** — n/a (single dimension), but *area given* works backwards: if **area** doubles, side ×√2, perimeter ×√2.
2. 🔴 **SSC shortcut:** Square scales purely by side. Area = side^2, so `Area factor = (side factor)²`. Perimeter & diagonal = (side factor)^1.
3. **Memory trick:** "*Square = the name says it — everything about area is squared.*" Side ×k ⇒ Area ×k^2.
4. **Exam pattern:** "Side of a square is increased by 20%, find % increase in area." (Answer 44%.) Or the reverse: "Area increased by 44%, find side increase" → √1.44 = 1.2 → **20%**.

5. 🟠 **Common mistakes:** (i) saying area ×2 when side ×2 (it's ×4). (ii) forgetting diagonal scales linearly (×k), not ×k^2.
6. **Numericals:**

- *Q.* Side doubles. Area? → **×4** ($4a^2$).
- *Q.* Side +20%, area increase? → $1.2^2 - 1$ = **44%**.
- *Q.* Perimeter increases 30%, area increases by? → side ×1.3 → area ×1.69 → **69%**.

1. **Revision:** Side ×k → **A ×k^2, P ×k, d ×k.**

A2. Rectangle



1. **Original formulas** — 🔵 Area `A = l·b` · Perimeter `P = 2(l + b)` · Diagonal `d = √(l² + b²)`
2. **One dimension changes:**

- Length ×2 (b fixed) → 🟢 `A = 2l·b` → **Area ×2**. Perimeter changes *non-uniformly* (only l part), so **no single factor** — must compute.
- Breadth ÷2 (l fixed) → Area ×½.

1. **Multiple dimensions:** Area factor = **(l-factor) × (b-factor)**.

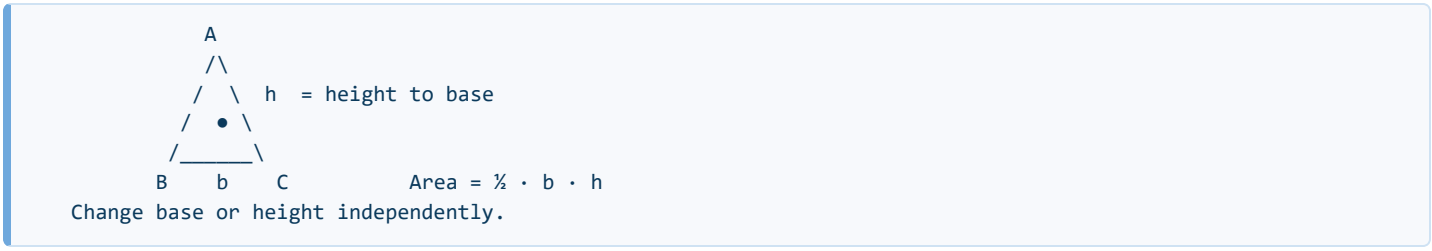
- Length ×2 **and** breadth ×½ → Area ×(2 × ½) = **×1 (unchanged)**. 🔴 classic trap answer.
- Length +20%, breadth +20% → 1.2×1.2 = **+44%**.
- Length +20%, breadth −20% → 1.2×0.8 = 0.96 → **−4%** (use `a+b+ab/100`).

1. 🔴 **SSC shortcut:** For rectangle **area only multiply the two factors**; perimeter does *not* follow a clean factor unless both change by the same k.
2. **Memory trick:** "*Two sides, two factors — multiply them for area.*"
3. **Exam pattern:** "Length doubled, breadth halved — effect on area?" (Unchanged). "L +10%, B +20%, area change?" (`10+20+2 = 32%`).
4. 🟠 **Mistakes:** (i) applying a single factor to perimeter when only one side changes. (ii) adding the two % for area instead of using `a+b+ab/100`. (iii) forgetting the cross-term (the `ab/100`).
5. **Numericals:**

- *Q.* L ×3, B ×2 → Area **×6**.
- *Q.* L −10%, B −20% → `-10 -20 + 2 = -28%` (area falls 28%).
- *Q.* L +50%, B unchanged → Area **+50%** (only one factor).

1. **Revision:** Area ×(k_l·k_b). If both ×k → A ×k^2, P ×k, d ×k.

A3. Triangle (general)



1. **Original formula** — 🔵 Area `A = ½ · b · h` · Perimeter = sum of sides.
2. **One dimension changes:**

- Base ×2 (h fixed) → 🟢 `A = ½(2b)h = b·h` → **Area ×2**.
- Height ÷2 → Area ×½.

1. **Multiple dimensions:** Area factor = **(base-factor) × (height-factor)**.

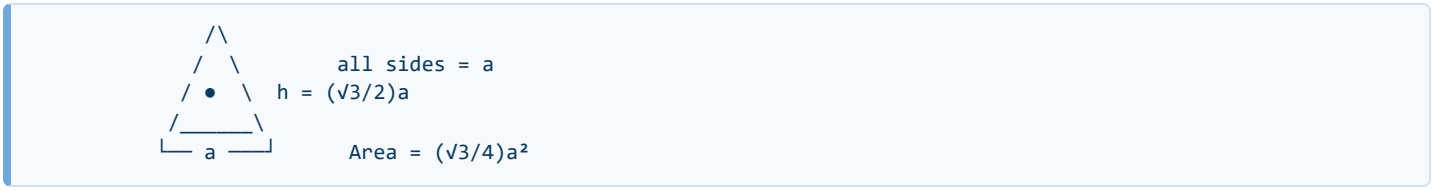
- Base +20%, height −25% → 1.2 × 0.75 = 0.9 → **−10%**.
- Base ×2, height ×3 → **×6**.

1. 🔴 **SSC shortcut:** Triangle area behaves exactly like rectangle — it's just ½ of one. Only base × height matter.
2. **Memory trick:** "*½ is a constant — it never changes the factor. Watch base and height only.*"
3. **Exam pattern:** "Base increased 20%, height decreased 20%, area change?" (`20 −20 −4 = −4%`).
4. 🟠 **Mistakes:** (i) thinking the ½ affects scaling (it cancels). (ii) using a *side* instead of the base⊥height pair.
5. **Numericals:**

- *Q.* Base ×4, height ÷2 → Area **×2**.
- *Q.* Base +30%, height +30% → `30+30+9 = 69%`.

1. **Revision:** A ×(k_b·k_h). All-sides ×k (similar triangle) → A ×k^2, P ×k.

A4. Equilateral Triangle



1. **Original formulas** — 🔵 Area `A = (√3/4)a²` · Perimeter `P = 3a` · Height `h = (√3/2)a`
2. **One dimension (side a):** it's the *only* dimension → behaves like a square.

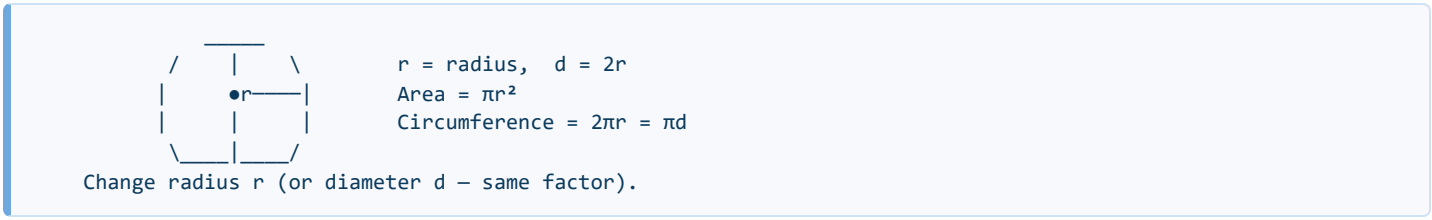
- Side ×2 → 🟢 `A = (√3/4)(2a)² = 4·(√3/4)a²` → **Area ×4**; P ×2; h ×2.
- Side ÷2 → Area ×¼.

1. **Multiple dimensions:** n/a — single dimension.
2. 🔴 **SSC shortcut:** Everything with a^2-form (equilateral triangle, square, circle) scales ×k^2 in area for side ×k.
3. **Memory trick:** "*Equal sides → treat like a square: area is a squared quantity.*"
4. **Exam pattern:** "Side of equilateral triangle tripled, area becomes?" (×9).
5. 🟠 **Mistakes:** (i) scaling height by k^2 (it's k). (ii) forgetting the √3/4 is a constant and cancels in ratios.
6. **Numericals:**

- *Q.* Side +100% (doubles) → Area **+300%** (×4).
- *Q.* Side +50% → Area 1.5^2 = 2.25 → **+125%**.

1. **Revision:** Side ×k → A ×k², P ×k, h ×k.

A5. Circle



1. **Original formulas** — 🔵 Area `A = πr²` · Circumference `C = 2πr = πd` · Diameter `d = 2r`
2. **One dimension (radius r):**

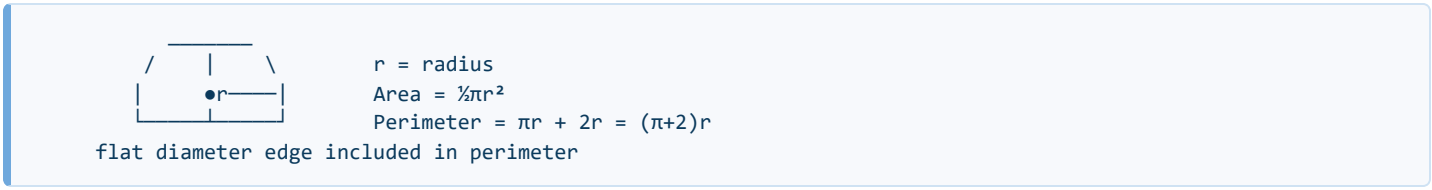
- Radius ×2 → 🟢 `A = π(2r)² = 4πr²` → **Area ×4**; `C = 2π(2r)` → **Circumference ×2**.
- Radius ÷2 → Area ×¼, Circumference ×½.
- **Diameter ×2 = Radius ×2** (same factor) → Area ×4, C ×2.

1. **Multiple dimensions:** n/a — a circle has one dimension.
2. 🔴 **SSC shortcut:** Circle = pure r^2 for area, r^1 for circumference. Diameter and radius scale identically.
3. **Memory trick:** "*Area is π-r-squared → the '-squared' tells you the power.*"
4. **Exam pattern:** "Radius increased 50%, area increases by?" (2.25 → **125%**). "Circumference doubles, area becomes?" (r ×2 → **×4**).
5. 🟠 **Mistakes:** (i) circumference ×4 when radius ×2 (it's ×2). (ii) treating a *diameter* change as different from radius — it's the same factor.
6. **Numericals:**

- *Q.* Radius halves → Area **¼**, Circumference **½**.
- *Q.* Diameter tripled → Area **×9**, Circumference **×3**.
- *Q.* Area increased 44% → radius √1.44 = ×1.2 → **+20%**.

1. **Revision:** r (or d) ×k → **A ×k², C ×k.**

A6. Semicircle



1. **Original formulas** — 🔵 Area `A = ½πr²` · Perimeter `P = πr + 2r = (π + 2)r` (curved arc **+** diameter).
2. **One dimension (r):**

- Radius ×2 → 🟢 `A = ½π(2r)² = 4·½πr²` → **Area ×4**; `P = (π+2)(2r)` → **Perimeter ×2**.
- Radius ÷2 → Area ×¼, Perimeter ×½.

1. **Multiple dimensions:** n/a.
2. 🔴 **SSC shortcut:** Same powers as a full circle — the ½ and (π+2) are constants that cancel in any ratio.
3. **Memory trick:** "*Half a circle scales exactly like a whole circle.*"
4. **Exam pattern:** "Radius of a semicircular field doubled — perimeter change?" (**×2**).
5. 🟠 **Mistakes:** (i) **forgetting the diameter** in perimeter — using only πr. Perimeter is `πr + 2r`, not `πr`. (ii) scaling perimeter by k^2.

6. **Numericals:**

- *Q.* r = 7, perimeter? → (π+2)·7 = (22/7 + 2)·7 = 22 + 14 = **36 cm**.
- *Q.* r ×3 → Area **×9**, Perimeter **×3**.

1. **Revision:** r ×k → A ×k^2, P ×k. *Perimeter = curved arc + diameter.*

A7. Sector of a Circle

```
          • centre
         /|\  θ = angle
        / | \
       /  |  \  r = radius
      /___|___\
       arc length ℓ = (θ/360)·2πr
Area = (θ/360)·πr²   ·   Perimeter = ℓ + 2r
```

1. **Original formulas** (angle θ) — 🔵 Area `A = (θ/360)πr²` · Arc `ℓ = (θ/360)·2πr = (θ/180)πr` · Perimeter `P = ℓ + 2r`.
2. **One dimension changes:**

- **Radius ×2** (θ fixed) → 🟢 Area `= (θ/360)π(2r)²` → **×4**; Arc `= (θ/360)2π(2r)` → **×2**.
- **Angle ×2** (r fixed) → Area ×2, Arc ×2 (θ is *linear* in both, since area's r is unchanged).

1. **Multiple dimensions:**

- Radius ×2 **and** angle ×2 → Area ×(4·2) = **×8**; Arc ×(2·2) = **×4**.
- Radius ÷2, angle ×2 → Area ×(¼·2) = **×½**; Arc ×(½·2) = **×1**.

1. 🔴 **SSC shortcut:** Treat **radius** with its normal powers (Area r^2, Arc r^1) and **angle θ as a plain linear multiplier** on both area and arc.
2. **Memory trick:** "*θ is a fraction of the circle → it scales area and arc equally and linearly. r keeps its square/linear powers.*"
3. **Exam pattern:** "Radius doubled, angle same — arc length change?" (**×2**). "Angle doubled, radius same — area change?" (**×2**).
4. 🟠 **Mistakes:** (i) scaling area by $θ^2$ (θ is linear, not squared). (ii) forgetting the two radii `2r` in perimeter.
5. **Numericals:**

- *Q.* r ×3, θ same → Area **×9**, Arc **×3**.
- *Q.* θ from 60° to 120° (×2), r same → Area & Arc both **×2**.

1. **Revision:** Area ×(k_r^2·$k_θ$), Arc ×(k_r·$k_θ$).

A8. Rhombus

```
          • (diagonals meet at 90°)
         /|\
   d₁   /  |  \        side a = ½√(d₁² + d₂²)
     /     |     \      Area = ½ · d₁ · d₂
     \     |     /      Perimeter = 4a
   d₂   \  |  /
         \|/
```

1. **Original formulas** — 🔵 Area `A = ½·d₁·d₂` (diagonals) · Perimeter `P = 4a` · Side `a = ½√(d₁²+d₂²)`.
2. **One dimension:**

- Diagonal d_1 ×2 (d_2 fixed) → 🟢 `A = ½(2d₁)d₂` → **Area ×2**.
- Side ×2 (a *similar* rhombus, both diagonals ×2) → Area ×4, Perimeter ×2.

1. **Multiple dimensions:** Area factor = **(d_1-factor)(d_2-factor)**.

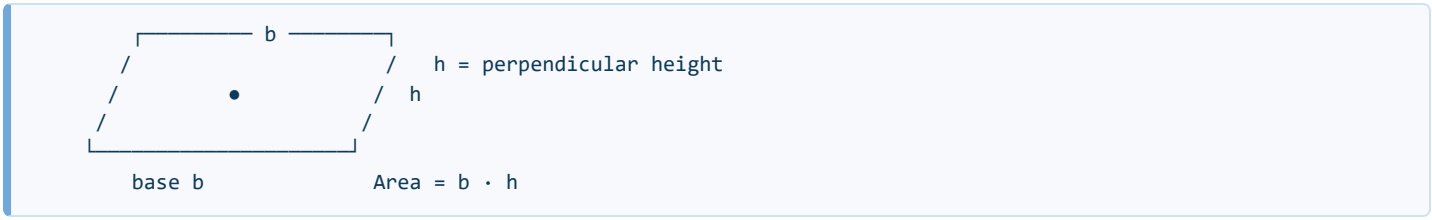
- Both diagonals ×2 → Area **×4**.
- d_1 +20%, d_2 +20% → **+44%**.

1. 🔴 **SSC shortcut:** Rhombus area works on **diagonals like a rectangle works on sides** — multiply the two factors.
2. **Memory trick:** "*Half-d-one-d-two — treat the two diagonals as the two 'sides' of area.*"
3. **Exam pattern:** "Both diagonals doubled, area?" (**×4**). "One diagonal +50%, other same?" (**+50%**).
4. 🟠 **Mistakes:** (i) using side for area (must use diagonals or side×height). (ii) forgetting the ½.
5. **Numericals:**

- *Q.* d_1 ×3, d_2 ÷2 → Area **×1.5**.
- *Q.* Side ×2 (similar) → Area **×4**, Perimeter **×2**.

1. **Revision:** Area ×(k_{d1}·k_{d2}); similar-scale side ×k → A ×k^2, P ×k.

A9. Parallelogram



1. **Original formula** — 🔵 Area `A = b · h` (base × ⊥ height) · Perimeter `P = 2(a + b)`.
2. **One dimension:**

- Base ×2 (h fixed) → 🟢 **Area ×2**.
- Height ÷2 → Area ×½.

1. **Multiple dimensions:** Area factor = **(base-factor)(height-factor)** (identical to rectangle/triangle logic).

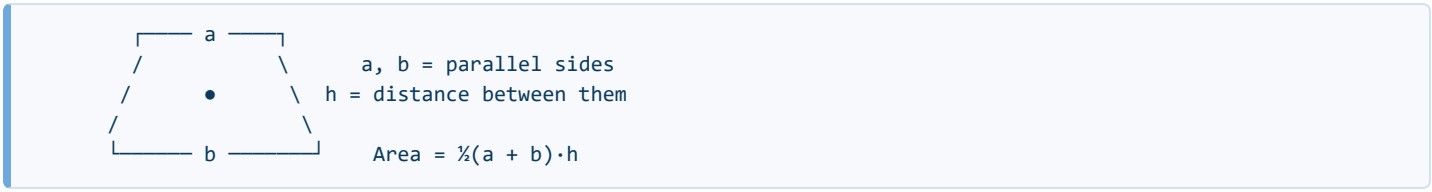
- Base ×2, height ×3 → **×6**. Base +25%, height −20% → 1.25×0.8 = **×1 (no change)**.

1. 🔴 **SSC shortcut:** Parallelogram = rectangle for scaling purposes — base × height only.
2. **Memory trick:** "*A slanted rectangle scales like a rectangle.*"
3. **Exam pattern:** "Base up 20%, height down 20%, area?" (`−4%`).
4. 🟠 **Mistakes:** (i) using the slant side instead of ⊥ height. (ii) same `a+b+ab/100` cross-term slip as rectangle.
5. **Numericals:**

- *Q.* Base ×4, height ÷2 → **×2**.
- *Q.* Base +10%, height +10% → **+21%**.

1. **Revision:** A ×(k_b·k_h).

A10. Trapezium



1. **Original formula** — 🔵 Area `A = ½(a + b)·h` (mean of parallel sides × height).

2. **One dimension:**

- Height ×2 (a, b fixed) → 🟢 `A = ½(a+b)(2h)` → **Area ×2**.
- A parallel side changing alone does **not** give a clean factor (it's `a+b`, a sum) — must compute.

1. **Multiple dimensions:**

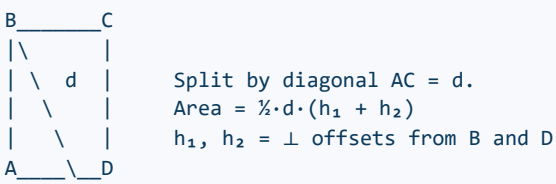
- **Both** parallel sides ×2 and height ×2 (similar figure) → Area **×4**.
- Height ×3, sides fixed → **×3**.

1. 🔴 **SSC shortcut:** Height is a clean linear factor. The `(a+b)` sum is **not** a factor unless both parallel sides scale by the *same* k.
2. **Memory trick:** "*Height is the honest factor; the two parallel sides add — you can't factor a sum.*"
3. **Exam pattern:** "Height of a trapezium doubled, parallel sides same — area?" (**×2**). Similar-scaled (all ×k) → area ×k^2.
4. 🟠 **Mistakes:** (i) treating `(a+b)` as multiplying when only one parallel side changes. (ii) forgetting the ½.
5. **Numericals:**

- *Q.* a = 6, b = 10, h = 4 → A = ½(16)(4) = **32 cm²**. Now h ×2 → **64 cm²** (×2). ✓
- *Q.* All linear dims ×3 (similar) → Area **×9**.

1. **Revision:** Height ×k → A ×k. All-linear ×k → A ×k^2.

A11. Quadrilateral (general, via a diagonal)



1. **Original formula** — 🔵 Area `A = ½·d·(h₁ + h₂)` (diagonal × sum of the two offsets).
2. **One dimension:**

- Diagonal d ×2 (offsets fixed) → 🟢 **Area ×2**.
- An offset alone → sum `(h₁+h₂)`, no clean factor.

1. **Multiple dimensions:** Fully similar (all lengths ×k) → **Area ×k^2**, perimeter ×k.
2. 🔴 **SSC shortcut:** Any planar figure, when *uniformly* scaled by k, has **area ×k^2, perimeter ×k** — no need for its specific formula.
3. **Memory trick:** "*Whatever the shape, uniform scale → area squares, boundary just scales.*"
4. **Exam pattern:** "A field's map is scaled up 3×; actual area is how many times the model?" (**×9**).
5. 🟠 **Mistakes:** scaling a *sum* term as if it were a single factor.
6. **Numericals:**

- *Q.* All sides & diagonals ×2 → Area **×4**.
- *Q.* Diagonal ×3, offsets same → Area **×3**.

1. **Revision:** Uniform ×k → A ×k^2, P ×k.

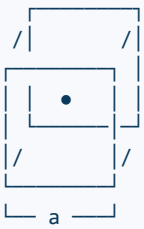
★ 2D Master Recap

Figure	Area formula	Uniform side/radius ×k ⇒ Area	⇒ Perimeter/Circ
Square	a^2	$\times k^2$	×k
Rectangle	l·b	$\times k^2$ (both), or $k_l \cdot k_b$	×k (both)
Triangle	½bh	$\times k^2$ (similar)	×k
Equilateral △	$(\sqrt{3}/4)a^2$	$\times k^2$	×k
Circle	πr^2	$\times k^2$	C ×k
Semicircle	$\frac{1}{2}\pi r^2$	$\times k^2$	×k
Sector	$(\theta/360)\pi r^2$	$\times k^2$ (r), ×linear (θ)	arc ×k
Rhombus	$\frac{1}{2}d_1d_2$	$\times k^2$ (both diag)	×k
Parallelogram	b·h	$\times k^2$ (similar)	×k
Trapezium	½(a+b)h	$\times k^2$ (similar)	×k
Quadrilateral	$\frac{1}{2}d(h_1+h_2)$	$\times k^2$ (similar)	×k

PART B — 3D SOLIDS

*In 3D: slant/diagonal/edge scale as **k**; every surface area (CSA/LSA/TSA) scales as **k^2**; volume scales as **k^3**.*

B1. Cube



```
edge = a
Volume = a³
TSA = 6a²
Space diagonal = a√3
```

1. **Original formulas** — 🔵 Volume `V = a³` · TSA `= 6a²` · LSA `= 4a²` · Space diagonal `= a√3` · Edge total `= 12a`.
2. **One dimension (edge a — the only one):**

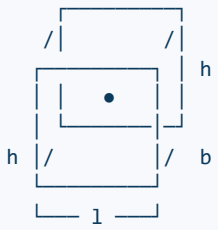
- Edge ×2 → 🟢 `V = (2a)³ = 8a³` → **Volume ×8**; TSA `= 6(2a)² = 24a²` → **×4**; diagonal ×2.
- Edge ÷2 → V ×⅛, SA ×¼, diagonal ×½.

1. **Multiple dimensions:** n/a — one dimension.
2. 🔴 **SSC shortcut:** Cube is the purest case: edge ×k → **V ×k^3, SA ×k^2, diagonal ×k.**
3. **Memory trick:** "*Cube → cubes it: 1-2-3 powers. Length k, Surface k^2, Volume k^3.*"
4. **Exam pattern:** "Edge of a cube increased 50%, volume increases by?" ($1.5^3 = 3.375$ → **237.5%**). "Volume increased 8×, edge?" (**×2**).
5. 🟠 **Mistakes:** (i) SA ×8 when edge ×2 (SA is ×4). (ii) diagonal ×k^2 (it's ×k).
6. **Numericals:**

- *Q.* Edge tripled → V **×27**, SA **×9**.
- *Q.* Edge +20% → V 1.728 → **+72.8%**, SA 1.44 → **+44%**.

1. **Revision:** Edge ×k → **V ×k³, TSA/LSA ×k², diagonal ×k.**

B2. Cuboid



```
l = length, b = breadth, h = height
V = l·b·h
TSA = 2(lb + bh + hl)
Space diagonal = √(l² + b² + h²)
```

1. **Original formulas** — 🔵 `V = lbh` · `TSA = 2(lb + bh + hl)` · `LSA = 2h(l + b)` · diagonal `= √(l²+b²+h²)`.
2. **One dimension:**

- Height ×2 (l, b fixed) → 🟢 `V = lb(2h)` → **Volume ×2**. TSA/diagonal have no clean single factor (mixed sums).

1. **Multiple dimensions:** Volume factor = **k_l · k_b · k_h**.

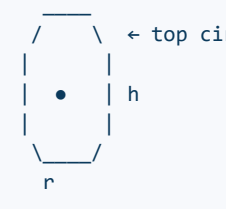
- **Length doubled, breadth tripled, height halved** → `2 × 3 × ½` = **Volume ×3**.
- All three ×2 (similar) → V ×8, TSA ×4, diagonal ×2.
- l +10%, b +20%, h −20% → 1.1×1.2×0.8 = 1.056 → **+5.6%**.

1. 🔴 **SSC shortcut: Volume = multiply the three factors.** TSA/diagonal only get a clean factor when *all* dimensions scale by the same k.
2. **Memory trick:** "*Three sides → three factors, multiplied, for volume.*"
3. **Exam pattern:** "L ×2, B ×3, H ÷2 — volume becomes?" (**×3**). "All edges +10% — volume up?" (1.331 → **33.1%**).
4. 🟠 **Mistakes:** (i) adding the three factors instead of multiplying. (ii) applying one factor to TSA when only one edge changes.
5. **Numericals:**

- *Q.* L ×2, B ×2, H ×2 → V **×8**.
- *Q.* L −20%, B −20%, H unchanged → V ×0.64 → **−36%** (an *area-type* change since one dim fixed).

1. **Revision:** V ×(k_l·k_b·k_h); uniform ×k → V ×k³, TSA ×k², diagonal ×k.

B3. Cylinder



```
V = πr²h
CSA = 2πrh
TSA = 2πr(r + h) = 2πrh + 2πr²
```

1. **Original formulas** — 🔵 `V = πr²h` · `CSA = 2πrh` · `TSA = 2πr(r + h)`.
2. **One dimension:**

- **Radius ×2** (h fixed) → 🟢 `V = π(2r)²h = 4πr²h` → **Volume ×4**; `CSA = 2π(2r)h` → **×2**.
- **Radius ÷2** → `V = π(r/2)²h = ¼πr²h` → **Volume ×¼**; CSA ×½.

- **Height ×2** (r fixed) → `V = πr²(2h)` → **Volume ×2**; `CSA = 2πr(2h)` → **×2**.
- **Height ÷2** → V ×½, CSA ×½.
- **Diameter ×2 = Radius ×2** → V ×4, CSA ×2.

1. **Multiple dimensions:** V factor = **(r-factor)² × (h-factor)**; CSA factor = **(r-factor) × (h-factor)**.

- **Radius ×2 & height ÷2** → V `= 4 × ½` = **×2**; CSA `= 2 × ½` = **×1 (unchanged)**.
- **Radius ÷2 & height ×2** → V `= ¼ × 2` = **×½**; CSA `= ½ × 2` = **×1**.
- **Radius ×2 & height ×2** (similar) → V ×8, CSA ×4, TSA ×4.

1. 🔴 **SSC shortcut:** Volume: **square the radius factor, times the height factor.** CSA: radius factor × height factor. TSA gets a single factor **only** if r and h scale together.
2. **Memory trick:** "*r is squared in volume, once in CSA; h is once in both.*"
3. **Exam pattern:** "Radius doubled, height halved — volume?" (**×2**). "Radius +50%, height same — CSA?" (**×1.5**).
4. 🟠 **Mistakes:** (i) squaring the height (h is linear). (ii) TSA — don't apply a single factor unless r and h scale equally; the r^2 and r·h terms scale differently otherwise.
5. **Numericals:**

- *Q.* r ×3, h same → V **×9**, CSA **×3**.
- *Q.* r same, h ×3 → V **×3**, CSA **×3**.
- *Q.* r +20%, h +50% → V = 1.44 × 1.5 = 2.16 → **+116%**.

1. **Revision:** V ×($k_r^2 \cdot k_h$), CSA ×($k_r \cdot k_h$); uniform ×k → V ×k^3, CSA/TSA ×k^2.

B4. Cone

```
      /\
     /  \   l = slant height = √(r² + h²)
  h /  • |     (h = vertical height)
   /_____|
   └── r ─┘      V = ⅓πr²h
                 CSA = πrl      TSA = πr(l + r)
```

1. **Original formulas** — 🔵 `V = ⅓πr²h` · `CSA = πrl` · `TSA = πr(l + r)` · Slant `l = √(r² + h²)`.
2. **One dimension:**

- **Radius ×2** (h fixed) → 🟢 `V = ⅓π(2r)²h = 4·⅓πr²h` → **Volume ×4**. (CSA changes non-trivially because *l* depends on r.)
- **Height ×2** (r fixed) → `V = ⅓πr²(2h)` → **Volume ×2**.
- **Radius ÷2** → Volume ×¼.
- **Slant height ×2** with r same → `CSA = πr(2l)` → **CSA ×2** (treating l as independent).

1. **Multiple dimensions:** V factor = **(r-factor)² × (h-factor)**.

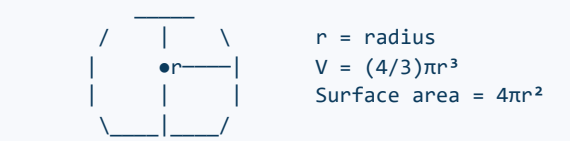
- **Radius ×2 & height halved** → V `= 4 × ½` = **×2**.
- **Radius ÷2 & height ×2** → V `= ¼ × 2` = **×½**.
- All linear (r, h, l) ×2 → V ×8, CSA ×4, TSA ×4.

1. 🔴 **SSC shortcut:** Cone volume scales **exactly like a cylinder** ($r^2 \cdot h$) — the ⅓ is a constant. For CSA/TSA, if the whole cone is scaled by k, they go ×k^2.
2. **Memory trick:** "*⅓ is a decoration — cone volume = cylinder volume rules ($r^2 h$).*"
3. **Exam pattern:** "Radius doubled, height halved — cone volume?" (**×2**). "Radius same, height tripled — volume?" (**×3**).
4. 🟠 **Mistakes:** (i) forgetting **l depends on r and h** — you can't hold CSA's factor clean when only r or only h changes. (ii) scaling CSA by r^2 (it's r·l).
5. **Numericals:**

- *Q.* r ×3, h same → V **×9**.
- *Q.* r same, h ×2 → V **×2**.
- *Q.* Whole cone scaled ×2 → V ×8, CSA ×4, slant ×2.

1. **Revision:** V ×(k_r²·k_h); uniform ×k → V ×k³, CSA/TSA ×k², slant ×k.

B5. Sphere



1. **Original formulas** — 🔵 `V = (4/3)πr³` · Surface `S = 4πr²` · Diameter `d = 2r`.
2. **One dimension (r):**

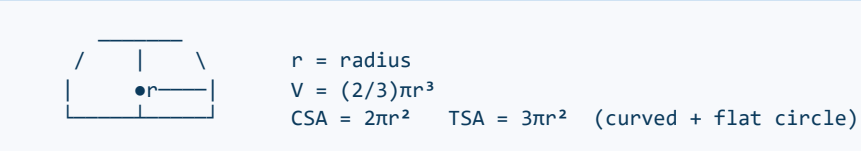
- **Radius ×2** → 🟢 `V = (4/3)π(2r)³ = 8·(4/3)πr³` → **Volume ×8**; `S = 4π(2r)²` → **Surface ×4**.
- **Radius ×3** → **Volume ×27**, **Surface ×9**.
- **Radius ÷2** → Volume ×⅛, Surface ×¼.
- **Diameter ×2 = Radius ×2** → V ×8, S ×4.

1. **Multiple dimensions:** n/a — one dimension.
2. 🔴 **SSC shortcut:** Sphere is the cleanest 3D case: **r ×k → V ×k³, S ×k².** No height, no slant.
3. **Memory trick:** "*Sphere = radius only. Volume cubes it, surface squares it.*"
4. **Exam pattern:** "Radius of a sphere increased 50%, volume increases by?" ($1.5^3 = 3.375$ → **237.5%**). "Surface ×9 → radius?" (**×3**).
5. 🟠 **Mistakes:** (i) V ×4 when r ×2 (it's ×8 — cube it). (ii) confusing diameter change as different from radius.
6. **Numericals:**

- *Q.* r ÷2 → V **×⅛**, S **×¼**.
- *Q.* r +20% → V 1.728 → **+72.8%**, S 1.44 → **+44%**.

1. **Revision:** r ×k → **V ×k³, S ×k².**

B6. Hemisphere



1. **Original formulas** — 🔵 `V = (2/3)πr³` · `CSA = 2πr²` · `TSA = 3πr²`.
2. **One dimension (r):**

- Radius ×2 → 🟢 `V = (2/3)π(2r)³` → **×8**; CSA & TSA `= …(2r)²` → **×4**.
- Radius ÷2 → V ×⅛, SA ×¼.

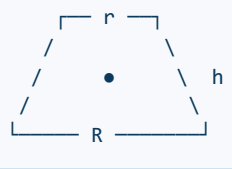
1. **Multiple dimensions:** n/a.
2. 🔴 **SSC shortcut:** Same powers as a sphere — r ×k → V ×k³, SA ×k². Constants (2/3, 2π, 3π) all cancel in ratios.
3. **Memory trick:** "*Half a ball, same power law: V cubes, SA squares.*"
4. **Exam pattern:** "Radius of a hemispherical bowl doubled — volume?" (**×8**).

5. 🟠 **Mistakes:** (i) forgetting TSA = $3\pi r^2$ includes the **flat circular face** (CSA is only $2\pi r^2$). (ii) scaling SA by k^3.
6. **Numericals:**

- *Q.* r ×3 → V **×27**, SA **×9**.
- *Q.* r +50% → V **+237.5%**, SA **+125%**.

1. **Revision:** r ×k → V ×k^3, CSA/TSA ×k^2.

B7. Frustum of a Cone




```
r = top radius, R = bottom radius
h = height,  l = √(h² + (R−r)²)
V = ⅓πh(R² + Rr + r²)
CSA = π(R + r)l
TSA = π(R + r)l + πR² + πr²
```

1. **Original formulas** — 🔵 `V = ⅓πh(R² + Rr + r²)` · `CSA = π(R + r)l` · `TSA = CSA + πR² + πr²` · Slant `l = √(h² + (R−r)²)`.
2. **One dimension:** because R, r, h are interlinked through sums, a *single* dimension change has **no clean factor** — compute directly.
3. **Multiple dimensions — the exam-usable case:** scale **all** linear dimensions (R, r, h, l) by the same k → 🟢 **V ×k^3, CSA ×k^2, TSA ×k^2, slant ×k.**

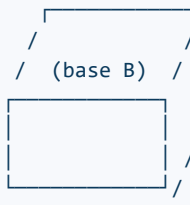
- Whole frustum ×2 → V ×8, CSA ×4.

1. 🔴 **SSC shortcut:** Don't try to factor a single dimension of a frustum. If it's *uniformly* scaled, apply $k^3/k^2/k$. Otherwise, plug numbers into the formula.
2. **Memory trick:** "*Frustum = a truncated cone; only whole-figure scaling gives clean k^3/k^2 powers.*"
3. **Exam pattern:** "A bucket (frustum) enlarged uniformly 3× — capacity becomes?" (**×27**).
4. 🟠 **Mistakes:** (i) treating (R^2 + Rr + r^2) as scalable by one radius. (ii) slant `l` uses `(R − r)`, **not** `(R + r)`.
5. **Numericals:**

- *Q.* Whole frustum ×2 → V **×8**, CSA **×4**.
- *Q.* R = 6, r = 3, h = 4 → l = √(16 + 9) = **5**; CSA = π(9)(5) = **45π**.

1. **Revision:** Uniform ×k → V ×k^3, CSA/TSA ×k^2, slant ×k.

B8. Prism (uniform cross-section)




```
Base = polygon of area B, perimeter P
V = B · h  (length of prism)
LSA = P · h
TSA = 2B + P·h
```

1. **Original formulas** — 🔵 `V = (Base area B) × (length h)` · `LSA = (Base perimeter P) × h` · `TSA = 2B + Ph`.
2. **One dimension:**

- **Length h ×2** (base fixed) → 🟢 `V = B(2h)` → **Volume ×2**; `LSA = P(2h)` → **×2**.
- **Base scaled ×m** (all base lengths ×m, h fixed) → B ×m^2, P ×m → V ×m^2, LSA ×m^2.

1. **Multiple dimensions:** V factor = **(base-area factor) × (length factor)** = **$m^2 \cdot k$** (base linear ×m, length ×k).

- Base doubled linearly (m=2) & length ×3 → V ×(4·3) = **×12**.
- Whole prism ×k → V ×k^3, all SA ×k^2.

1. 🔴 **SSC shortcut:** Prism volume = base-area factor × length factor. Base area scales ×(base-linear)^2.
2. **Memory trick:** "*A prism is an extruded 2D shape — the base carries the 2D power law, the length adds one more.*"
3. **Exam pattern:** "Cross-section sides doubled, length same — volume?" (base area ×4 → **V ×4**).
4. 🟠 **Mistakes:** (i) scaling base area linearly (it's squared in base sides). (ii) forgetting `2B` in TSA.
5. **Numericals:**

- *Q.* Length ×5, base fixed → V **×5**.
- *Q.* Base linear ×2, length ×2 (similar) → V **×8**.

1. **Revision:** V ×(m^2·k); uniform ×k → V ×k^3, SA ×k^2.

B9. Pyramid

```
        ▲ apex
       /|\
      / | \  slant l
     /  h  \        V = ⅓ · (Base area B) · h
    /___|___\       LSA = ½ · (Base perimeter P) · l
   └── base ──┘     TSA = B + ½·P·l
```

1. **Original formulas** — 🔵 `V = ⅓ · B · h` · `LSA = ½ · P · l` · `TSA = B + ½Pl` · slant `l` (from h and base).
2. **One dimension:**

- **Height h ×2** (base fixed) → 🟢 `V = ⅓B(2h)` → **Volume ×2**.
- **Base scaled linearly ×m** (h fixed) → B ×m^2 → **Volume ×m^2**.

1. **Multiple dimensions:** V factor = **(base-area factor) × (height factor)** = **m^2 · k**.

- Base sides ×2 & height ×3 → V ×(4·3) = **×12**.
- Whole pyramid ×k → V ×k^3, LSA/TSA ×k^2, slant ×k.

1. 🔴 **SSC shortcut:** Pyramid volume = ⅓ · base-area factor · height factor; base area scales as (base-linear)^2. (Same law as prism, with the ⅓.)
2. **Memory trick:** "*⅓ · base · height — base is a 2D area (squares), height is linear.*"
3. **Exam pattern:** "Base edge doubled, height same — volume of a square pyramid?" (base area ×4 → **×4**).
4. 🟠 **Mistakes:** (i) scaling base area linearly. (ii) confusing slant height *l* with vertical height *h* in the volume formula (volume uses **h**, LSA uses **l**).
5. **Numericals:**

- *Q.* Height tripled, base same → V **×3**.
- *Q.* Whole pyramid ×2 → V **×8**, LSA **×4**, slant **×2**.

1. **Revision:** V ×(m^2·k); uniform ×k → V ×k^3, SA ×k^2, slant ×k.

PART C — ★ DIMENSION CHANGE SHORTCUT TABLE (Master)

🔴 *The exam's whole point in one board. "×k" reads "multiplied by k".*

2D figures *(4th column = perimeter / circumference / arc / diagonal)*

Figure	Change	Area	Boundary / other length
Square	Side ×2	×4	Perimeter ×2, diagonal ×2
Square	Side ×3	×9	Perimeter ×3, diagonal ×3
Rectangle	L ×2, B ×½	×1 (same)	compute (mixed)
Rectangle	L ×2, B ×3	×6	compute (mixed)
Triangle	Base ×2, Height ×3	×6	compute (mixed)
Equilateral △	Side ×2	×4	Perimeter ×2, height ×2
Circle	Radius ×2	×4	Circumference ×2
Circle	Radius ×3	×9	Circumference ×3
Circle	Radius ×½	×¼	Circumference ×½
Semicircle	Radius ×2	×4	Perimeter ×2
Sector	Radius ×2 (θ same)	×4	Arc ×2
Sector	Angle ×2 (r same)	×2	Arc ×2
Rhombus	Both diagonals ×2	×4	Perimeter ×2
Parallelogram	Base ×2, Height ×½	×1 (same)	compute (mixed)
Trapezium	Height ×2 (sides same)	×2	—

3D solids *(surface column = CSA/LSA; TSA scales as k^2 only under uniform scaling)*

Solid	Change	Volume	Surface (CSA/LSA/TSA)
Cube	Side ×2	×8	×4 (diagonal ×2)
Cube	Side ×3	×27	×9 (diagonal ×3)
Cuboid	L×2, B×3, H×½	×3	compute (mixed)
Cuboid	all ×2	×8	TSA ×4 (diagonal ×2)
Cylinder	Radius ×2 (h same)	×4	CSA ×2, TSA changes*
Cylinder	Height ×2 (r same)	×2	CSA ×2, TSA changes*
Cylinder	R ×2 & H ×½	×2	CSA ×1 (same), TSA changes*
Cylinder	all ×2	×8	CSA ×4, TSA ×4
Cone	Radius ×2 (h same)	×4	l changes — compute
Cone	Height ×2 (r same)	×2	l changes — compute
Cone	R ×2 & H ×½	×2	l changes — compute
Cone	all ×2	×8	CSA ×4, TSA ×4 (slant ×2)
Sphere	Radius ×2	×8	Surface ×4
Sphere	Radius ×3	×27	Surface ×9
Sphere	Radius ×½	×⅛	Surface ×¼
Hemisphere	Radius ×2	×8	CSA/TSA ×4
Frustum	all ×2	×8	CSA ×4 (slant ×2)
Prism	Length ×2 (base same)	×2	LSA ×2
Prism	Base linear ×2, len same	×4	compute
Pyramid	Height ×2 (base same)	×2	—
Pyramid	all ×2	×8	LSA ×4 (slant ×2)

\Cylinder/cone TSA gets a single clean factor **only** when r and h scale by the same k (then ×k^2); otherwise compute term-by-term.*

PART D — SSC SHORTCUTS (the reasoning behind every row)

🔴 **Shortcut 1 — Power counting.** Volume depends on **cubic** dimensions (k^3), all areas/surfaces on **square** dimensions (k^2), all length quantities (perimeter, circumference, arc, slant, diagonal, radius) change **linearly** (k^1).

🔴 **Shortcut 2 — Multiply the factors.** When several dimensions change together, multiply their individual factors, **each raised to the power it carries** in the formula.

`V_cylinder = πr²h` → radius ×a, height ×b → volume × **a^2b**.

🔴 **Shortcut 3 — Constants vanish.** π, ½, ⅓, √3/4, 4/3, (π+2) etc. are constants — they **cancel** in every "how many times / what %" question. Ignore them; only variable dimensions scale.

🔴 **Shortcut 4 — Percentages.** Convert to a factor `k = 1 ± x/100` first, apply the power, then subtract 1 for the % change. For two area-factors use `a + b + ab/100`.

🔴 **Shortcut 5 — Reverse questions.** "Volume became 8×, find edge change" → take the cube root of 8 = 2 → edge ×2 (+100%). Area ×k^2 given → side = √k^2. Volume ×k^3 given → side = $\sqrt[3]{k^3}$.

🔴 **Shortcut 6 — Diameter = Radius factor.** For circle/sphere/cylinder/cone, a diameter change is the **same factor** as the radius change — never double-count.

PART E — MEMORY TRICKS

```
┌──────────────────────────────────────────────┐
│   POWER LADDER  (the heart of the chapter)   │
│                                              │
│     LENGTH   →  1 power   →  ×k              │
│     AREA     →  2 power   →  ×k²             │
│     VOLUME   →  3 power   →  ×k³             │
│                                              │
│   Dimension ×2  →  L×2 · A×4 · V×8   (1-2-3 → 2-4-8)
│   Dimension ×3  →  L×3 · A×9 · V×27  (1-2-3 → 3-9-27)
│   Dimension ×½  →  L×½ · A×¼ · V×⅛           │
└──────────────────────────────────────────────┘
```

- **"L-A-V = 1-2-3."** Length, Area, Volume carry powers 1, 2, 3. Say it before every question.
- **"Double → 2-4-8, Triple → 3-9-27."** Memorise these two rows cold; most SSC scaling questions are one of them.
- **"Halve → ¼ area, ⅛ volume."** The tank/bucket favourite.
- **"+x% → (1 + x/100), then power."** Never apply % to area/volume directly.
- **"Two things change? Multiply — don't add."** (Except the `a + b + ab/100` % form, which *is* the multiplied factor expanded.)
- **"Constants cancel."** π, ½, ⅓ never affect a ratio.
- **Cylinder/Cone chant:** "*r-squared, h-once.*" Volume = r^2h-behaviour; CSA = r·(h or l).
- **Sphere chant:** "*radius only — cube for volume, square for skin.*"

PART F — TOP COMMON MISTAKES (🟠 avoid to score full marks)

#	Mistake 🟠	Correct thinking 🟢
1	Side +20% ⇒ area +20%	Area +44% (raise 1.2 to power 2 first)
2	Radius ×2 ⇒ sphere volume ×4	Volume **×8** (cube it)
3	Adding two % changes for area	Use `a + b + ab/100`
4	Applying one factor to cuboid TSA when only one edge changes	TSA is a sum — compute term-by-term
5	Cylinder TSA ×k when only r changes	Only clean if r **and** h scale equally
6	Scaling diagonal / slant by k^2	They're **length** → ×k
7	Counting diameter and radius as separate changes	Same factor — once
8	Forgetting the diameter in a semicircle's perimeter	P = πr **+ 2r**
9	Using slant `l` in cone/pyramid **volume**	Volume uses vertical height **h**; l is for CSA
10	Treating a *sum* term ($a+b$, R^2+Rr+r^2) as a single scalable factor	Only whole-figure uniform scaling gives clean powers

---

PART G — 📜 SSC LAST-MINUTE REVISION SHEET (2 pages)

🔵 Original · 🟢 New/Result · 🔴 Shortcut · 🟠 Mistake. Read this the night before.

PAGE 1 — The Law + 2D

```
THE ONE LAW:  scale every linear dim by k  →  Length ×k · Area ×k² · Volume ×k³
Percentage:   k = 1 ± x/100 ,  then raise to the power ,  then −1 for the %.
Two dims:     area % = a + b + (ab)/100   ·   volume factor = multiply each (to its power)
```

🔵 2D Formula	🔴 Uniform side/radius ×k	Key single-dim case
Square `a²` , `4a` , `a√2`	A ×k^2, P ×k, d ×k	side ×2 → A ×4
Rectangle `lb` , `2(l+b)`	A ×k^2, P ×k	L×2,B×½ → A same
Triangle `½bh`	A ×k^2	base×2,h×3 → A ×6
Equilateral `(√3/4)a²`	A ×k^2, P ×k	side ×3 → A ×9
Circle `πr²` , `2πr`	A ×k^2, C ×k	r ×2 → A ×4, C ×2
Semicircle `½πr²` , `(π+2)r`	A ×k^2, P ×k	🟠 P = πr **+2r**
Sector `(θ/360)πr²`	A ×k_r^2·k_θ	arc ×k_r·k_θ
Rhombus `½d₁d₂`	A ×k^2	both diag ×2 → A ×4
Parallelogram `bh`	A ×k^2	b×2,h×½ → A same
Trapezium `½(a+b)h`	A ×k^2	h ×2 → A ×2

2D quick numbers: ×2 → area ×4 · ×3 → area ×9 · ×½ → area ×¼ · +20% → area +44% · +50% → area +125%.

PAGE 2 — 3D Solids

```
SURFACE (CSA/LSA/TSA) ×k²   ·   VOLUME ×k³   ·   SLANT & DIAGONAL ×k
Cylinder/Cone volume:  r²·h behaviour  →  radius×a, height×b  ⇒  V ×a²b
```

🔵 3D Formula	🔴 Uniform ×k	Signature exam case 🟢
Cube `a³`, `6a²`, `a√3`	V ×k^3, SA ×k^2, d ×k	edge ×2 → V ×8, SA ×4
Cuboid `lbh`, `2(lb+bh+hl)`	V ×k^3, SA ×k^2	L×2,B×3,H×½ → V ×3
Cylinder `πr²h`, CSA `2πrh`	V ×k^3, CSA ×k^2	r×2,h×½ → V ×2, CSA same
Cone `⅓πr²h`, CSA `πrl`	V ×k^3, CSA ×k^2	r×2,h×½ → V ×2
Sphere `(4/3)πr³`, `4πr²`	V ×k^3, S ×k^2	r ×2 → V ×8, S ×4
Hemisphere `(2/3)πr³`, `3πr²`	V ×k^3, SA ×k^2	r ×2 → V ×8
Frustum `⅓πh(R²+Rr+r²)`	V ×k^3, CSA ×k^2	whole ×2 → V ×8
Prism `B·h`, LSA `P·h`	V ×k^3, SA ×k^2	length ×2 → V ×2
Pyramid `⅓Bh`, LSA `½Pl`	V ×k^3, SA ×k^2	height ×2 → V ×2

3D quick numbers: dim ×2 → V ×8, SA ×4 · ×3 → V ×27, SA ×9 · ×½ → V ×⅛, SA ×¼ · +50% → V +237.5% · −20% → V −48.8%.

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FINAL 10-SECOND CHECK before you tick an answer:
1. Is the quantity length (¹), area (²), or volume (³)?
2. Did I raise EACH factor to its own power?
3. Did I MULTIPLY (not add) the factors?
4. For %: did I convert to a factor and subtract 1 at the end?
5. Diameter = radius factor (count once). Constants cancelled.
```

✅ **Master this one page and every "if the dimension changes..." question in Geometry & Mensuration is a guaranteed mark.**